



Evaluating Large Language Models in Data Generation for Low-Resource Scenarios: A Case Study on Question Answering

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Abstract

Large Language Models (LLMs) are powerful tools for generating synthetic data, offering a promising solution to data scarcity in low-resource scenarios. This study evaluates the effectiveness of LLMs in generating question-answer pairs to enhance the performance of question answering (QA) models trained with limited annotated data. While synthetic data generation has been widely explored for text-based QA, its impact on spoken QA remains underexplored. We specifically investigate the role of LLM-generated data in improving spoken QA models, showing performance gains across both text-based and spoken QA tasks. Experimental results on subsets of the SQuAD, Spoken SQuAD, and a Turkish spoken QA dataset demonstrate significant relative F1 score improvements of 7.8%, 7.0%, and 2.7%, respectively, over models trained solely on restricted human-annotated data. Furthermore, our findings highlight the robustness of LLM-generated data in spoken QA settings, even in the presence of noise.

Index Terms: spoken question answering, large language models, data generation

1. Introduction

Recent advancements in dataset augmentation techniques have seen the integration of LLMs to enhance the quality and diversity of training data. LLMs are prompted to generate synthetic data, which can then be utilized to fine-tune a model, eliminating the necessity for manual data collection in large amounts. This approach not only reduces the time and cost associated with data curation but also allows for the generation of highly customized datasets tailored to specific tasks.

LLMs have proven to be effective synthetic data generators across diverse domains [1], including commonsense reasoning [2], information retrieval [3], code-switching Automatic Speech Recognition (ASR) [4], and QA [5, 6, 7]. While LLMs help address data scarcity in many tasks, their use for dataset augmentation also presents challenges, such as introducing factual inconsistencies through synthetic examples. To address these challenges, hybrid approaches that combine human expertise with LLM-driven annotation have been proposed to optimize data annotation processes [8, 9].

In this paper, we explore the potential of LLMs for QA data generation, particularly in low-resource scenarios and spoken QA tasks. Question generation has been shown to significantly enhance QA model performance [10, 11, 12, 13]. Traditional question generation approaches often involve fine-tuning generative language models [14, 15, 16], which typically require

extensive annotated data. However, in scenarios with limited or no training data, alternative strategies include cross-lingual knowledge transfer [17, 18, 19], few-shot learning [20], effective fine-tuning strategies [21] and leveraging LLMs for synthetic data generation [5, 6, 7].

Our research utilizes LLMs to distill knowledge into a QA model trained with limited annotated data. Specifically, we generate question-answer pairs from provided passages using LLMs and integrate this synthetic data to enhance both text-based and spoken QA models. Unlike approaches that rely on deploying large-scale LLMs for direct downstream tasks, our method demonstrates that small, efficient models can achieve substantial performance improvements when augmented with LLM-generated synthetic data. This finding is particularly relevant for applications with limited computational resources. To simulate low-resource scenarios, we employ small partitions from SQuAD [22] and Spoken SQuAD [23], as well as a low-resource Turkish spoken QA dataset. While synthetic data generation has been widely explored for text-based QA, its impact on spoken QA remains under-researched [21]. Our study systematically investigates how LLM-generated data can enhance spoken QA models, demonstrating significant performance gains even in presence of ASR noise. Furthermore, by focusing on a less-supported language, i.e., Turkish, we highlight the potential of LLMs to handle multilingual challenges and improve QA performance in low-resource languages.

Our work distinguishes itself from prior studies that utilized LLMs for QA data generation, which primarily focused on few-shot learning approaches [5, 6] or on creating spoken QA datasets from acoustic data [7]. Instead, we use off-the-shelf models, investigate their data generation performance for spoken QA as well as text-based QA and evaluate their effectiveness on publicly available and non-public datasets to mitigate the potential risk of data leakage [1] – an important concern when assessing LLMs on publicly available datasets. While our research shares a similar emphasis on enhancing spoken QA performance as in [21], our primary focus is on utilizing LLMs for data generation and extending this capability to low-resource languages beyond English.

2. Method

We employed LLMs to generate question-answer pairs for QA tasks. The data generation framework is illustrated in Figure 1. Question-answer generation follows a two-stage process. In Stage 1, the LLM receives a paragraph as input and produces question-answer pairs. In this stage LLM is tasked with producing a specific number of questions (represented as “NBR” in Figure 1) along with their corresponding answers from the given paragraph. Since we are generating data for reading/listening

This work was conducted while Yusufcan Manav was with Bogazici University.

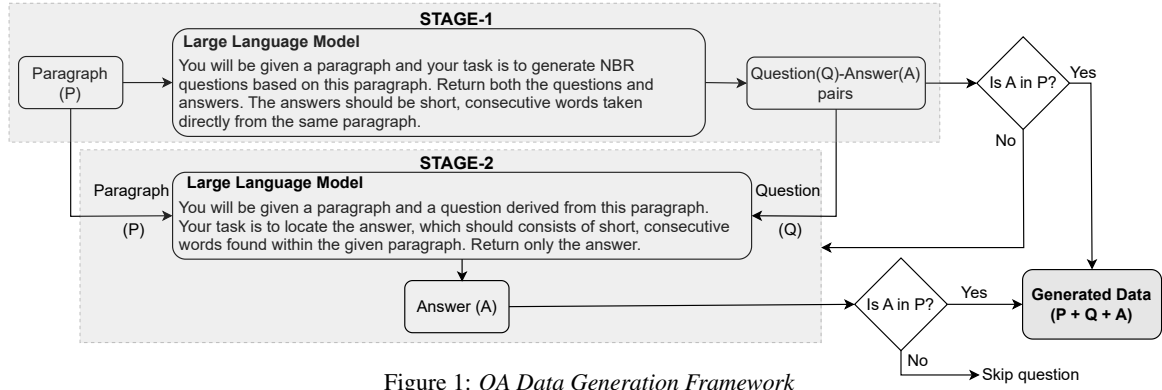


Figure 1: QA Data Generation Framework

comprehension tasks, the answers are expected to be short, consecutive words within the same paragraph. However, as LLM prompting does not consistently produce the desired output, we may not always obtain the expected number of questions for the given paragraph, and the answer may not necessarily be a span of words within the paragraph. To address the former issue, no specific action is taken, potentially leading to a reduced number of questions. To tackle the latter issue, we introduced Stage-2 in data generation. After obtaining the question-answer pairs from the LLM, we verify if the answer is indeed a consecutive span of words within the paragraph. If this criterion is met, the question-answer pair is included in the generated data along with the corresponding paragraph. This verification process is applied to all generated questions. If the answer fails to meet the criteria, the process proceeds to Stage-2.

In Stage-2, the LLM is provided with the same paragraph as in Stage-1, along with the question whose answer did not pass the verification process. In this stage LLM is prompted to locate the answer within the given paragraph. The generated answer undergoes the same verification process as in Stage-1. If the answer meets this criterion, the question-answer pair is included in the generated data for the given paragraph. However, if the answer does not satisfy the verification process, the question is skipped. As a result, we may end up with fewer question-answer pairs than desired. This 2-state procedure is applied to all paragraphs in the unannotated data.

The prompts used to invoke the LLM are illustrated in Figure 1. These prompts are specifically designed for question generation from clean text. When generating questions from ASR transcriptions, the prompts are adapted to inform the LLM about the potential presence of inaccuracies due to speech recognition errors.

3. Experimental Setup

This section explains the data, models and setups utilized in both question-answer generation and QA tasks.

3.1. Data

In our experiments, we used SQuAD [22], Spoken SQuAD [23] and Turkish lecture [24]¹ QA datasets. SQuAD contains 100,000+ question-answer pairs across 23.2K paragraphs for training, development, and test partitions. Spoken SQuAD contains ASR transcriptions of SQuAD paragraphs, along with corresponding question-answer pairs. Spoken SQuAD excludes pairs where the answer couldn't be found in the ASR transcriptions, resulting in approximately 37K and 5.3K question-answer

pairs for training and development sets. Additionally, the development set is available in three versions – No Noise, Noise V1, and Noise V2 – each with different word error rates (WERs) of 22.7%, 44.2%, and 54.8%, respectively. The Turkish lecture dataset was collected from lecture recordings of law courses. It consists of approximately 18 hours of video recordings and their corresponding reference and ASR transcriptions. The transcriptions were segmented into short passages and manually annotated to generate QA pairs, resulting in 878, 286, and 287 pairs for the training, development, and test partitions, respectively. Unlike the SQuAD datasets, this dataset contains longer answers, averaging 15 words. Turkish lecture dataset is also used for spoken QA by utilizing the ASR transcriptions of the lecture recordings.

The Turkish lecture dataset represents a typical low-resource QA scenario. For data generation, we utilized reference transcriptions from approximately 12 hours of lecture recordings within the same domain. Since the transcriptions lacked predefined paragraph boundaries, we created pseudo-paragraphs by segmenting the text into fixed-length chunks, which served as unannotated Turkish data. To simulate low-resource conditions for the (spoken) SQuAD datasets, we prepared annotated data subsets containing 1K, 3K, and 5K question-answer pairs, with each larger subset including the smaller ones. For LLM-based QA generation, we selected raw paragraph subsets that were disjoint from the annotated data, aiming to generate approximately 1K, 5K, and 10K question-answer pairs – matching the sizes of the original datasets.

3.2. Question-Answer Generation

To generate QA pairs from the SQuAD datasets using LLMs, we employed four models: GPT-4o-mini [25, 26], Llama-3.2-3B [27], Llama-3-8B [27], and Qwen2.5-7B [28]. All models were run on an NVIDIA GeForce RTX 4090 GPU. Due to computational constraints, we limited our experiments to models with at most 8B parameters. We followed the framework depicted in Figure 1 for each model. Each paragraph was used to prompt the LLM to generate a number of questions equivalent to those in the original dataset, ensuring the generation of 1K, 5K, and 10K synthetic question-answer pairs. The prompts primarily followed the structure shown in the figure. Due to cost and response limitations, we used GPT-4o-mini only for generating 1K and 5K question-answer pairs. The 10K question-answer pairs were generated exclusively using the open-source LLMs.

Although Turkish is not officially supported by all the open-source LLMs mentioned, we used them to assess their multilingual capabilities and performance on a non-public dataset. Since data leakage is a potential concern when evaluating LLMs

¹This dataset is non-public and was used with permission.

Table 1: Answer Relevancy Scores

Metrics	English (1K generated examples)			Turkish (~1.5K generated examples)		
	Llama-3.2-3B	Llama-3-8B	Qwen2.5-7B	Llama-3.2-3B	Llama-3-8B	Qwen2.5-7B
Answer Relevancy	0.64	0.65	0.63	0.21	0.24	0.24

on publicly available datasets [1], using a non-public dataset of offers a more reliable assessment.

3.3. Question Answering

For the QA task in English, we utilized pretrained encoder-only models such as BERT [29] and its variants (RoBERTa, ALBERT [30]), as well as ELECTRA [31]. For the QA task in Turkish, we employed BERTurk and the multilingual XLM-RoBERTa [32]. These models were fine-tuned using both the annotated data subsets and synthetically generated QA data. Following the findings of [21], we adopted a two-step fine-tuning strategy with the generated data. In this approach, the QA model is first fine-tuned on the generated data and then further fine-tuned on the manually annotated data. This strategy has the advantage of preventing the model from predominantly learning repetitive or inaccurate questions and answers that may arise from automatically generated data.

The QA system was implemented using the HuggingFace transformers library [33]. During fine-tuning, we set the batch size to 4 and experimented with learning rates of 2e-5 and 3e-5. The best epoch was selected based on F1 performance. Since the test partition of SQuAD is not publicly available, we evaluated the QA performance on the official SQuAD and Spoken SQuAD development sets. Hyperparameters were tuned on a separate partition selected from the training set.

4. Results

While LLMs offer a convenient method for generating synthetic data, they may introduce biases and factual inconsistencies in question-answer pairs. Therefore, assessing the quality of the automatically generated data is essential. In this study, we evaluated the quality of generated examples using both direct and indirect approaches. Although human evaluators provide reliable direct assessments, the process is costly and time-consuming. To mitigate this, we also leveraged an LLM as an evaluator for direct assessment. For indirect assessment, we utilized the generated data in the downstream QA task and evaluated the effectiveness of the generated data.

4.1. Direct Evaluation of the Generated Data

To evaluate the automatically generated question-answer pairs, we used GPT-4o-mini as an evaluator for data generated by Llama and Qwen LLMs. Since only raw paragraphs were used for data generation and the original question-answer pairs were excluded, we employed metrics that do not require ground truth, such as faithfulness and answer relevancy, to assess the quality of the LLM-generated examples. Faithfulness measures the factual consistency of an answer with its context, while answer relevancy evaluates how well an answer aligns with the generated question. Due to the nature of reading comprehension task, faithfulness scores close to 1.0 are expected. As a result, we focused on evaluating answer relevancy by generating a set of artificial questions based on the response. These questions were designed to reflect the content of the response, and the answer relevancy score was computed as the average cosine similarity between the embeddings of the input question and the gener-

Table 2: F1 results on SQuAD with 1K - 10K generated data with ALBERT-large / ALBERT-xxlarge models.

#QA Pairs	#Gen Pairs	GPT-4o mini	Llama-3.2 3B	Llama-3 8B	Qwen2.5 7B
1K	0	78.0 / 89.2 (Baseline)			
0	1K	72.7 / 84.7	72.0 / 84.0	73.0 / 86.8	75.7 / 86.7
0	5K	77.4 / 85.9	77.3 / 86.9	79.1 / 88.2	81.2 / 89.9
0	10K	- / -	78.6 / 88.3	79.3 / 89.0	82.9 / 90.9
1K	1K	79.4 / 89.3	79.1 / 89.6	79.6 / 89.2	80.3 / 90.0
1K	5K	80.7 / 90.3	81.3 / 90.1	82.3 / 90.3	81.8 / 91.2
1K	10K	- / -	82.6 / 90.7	83.6 / 91.6	84.1 / 91.9
zero-shot		90.5	81.2	86.2	91.8

ated questions. For evaluation, we utilized the Ragas library [34] with the generated question-answer pairs from SQuAD and Turkish dataset. The results, presented in Table 1, indicate that the generated pairs for English are more relevant than those for Turkish, suggesting higher data quality in English.

4.2. English QA Results

English experiments were performed by fine-tuning the encoder-only models described in Section 3.3 using both annotated and LLM-generated data subsets.

4.2.1. QA Results

For the baseline English QA systems, we fine-tuned the encoder-only models described in Section 3.3 using only annotated data subsets. The highest F1 score was achieved with the ALBERT-xxlarge model, which has approximately 206M parameters. To evaluate the impact of LLM-generated data on smaller models, we also experimented with the ALBERT-large model, containing around 17M parameters. While larger models generally outperform smaller ones, smaller models offer benefits such as reduced computational costs and greater feasibility for deployment in resource-limited environments. After fine-tuning with just 1K annotated samples from the training partitions, ALBERT-large and ALBERT-xxlarge achieved F1 scores of 78.0 and 89.2 on SQuAD (see Table 2), and 64.0 and 77.6 on Spoken SQuAD (see Table 3), respectively.

Table 2 presents the baseline F1 scores, the F1 scores of ALBERT models fine-tuned with question-answer pairs generated by GPT-4o-mini, Llama, and Qwen LLMs, and the F1 scores of models fine-tuned with a combination of 1K annotated data and generated data. Additionally, the table includes the zero-shot performance of these LLMs on SQuAD. Columns with two F1 values correspond to ALBERT-large and ALBERT-xxlarge, respectively. The results indicate that performance improves as more generated data is incorporated. With only 10K synthetic examples, the models match or slightly exceed baseline performance. Combining 1K annotated data with generated data further boosts performance, particularly for ALBERT-large, achieving a 7.8% relative improvement (78.0 $\rightarrow$ 84.1). Among the LLMs, Qwen2.5-7B demonstrates the highest zero-shot performance, while the fine-tuned ALBERT-xxlarge model surpasses the zero-shot performance of the Llama models.

Table 3: *F1 results on Spoken SQuAD with 1K - 10K generated data with ALBERT-large / ALBERT-xxlarge models.*

#QA Pairs	#Gen Pairs	GPT-4o mini	Llama-3.2 3B	Llama-3 8B	Qwen2.5 7B
1K	0	64.0 / 77.6 (Baseline)			
1K	1K	64.8 / 78.6	64.6 / 78.5	65.1 / 78.4	65.8 / 77.8
1K	5K	66.2 / 77.9	67.2 / 78.0	68.3 / 80.1	68.1 / 79.5
1K	10K	– / –	67.3 / 80.1	68.5 / 80.3	68.2 / 80.1
zero-shot		–	66.6	72.9	77.8

Table 3 presents the F1 results on Spoken SQuAD. Compared to SQuAD, performance drops significantly on spoken data. However, LLM-based data generation remains effective, yielding relative improvements of 7.0% (64.0 → 68.5) for ALBERT-large and 3.5% (77.6 → 80.3) for ALBERT-xxlarge. The reported results are based on generated data from SQuAD. To examine the impact of ASR transcriptions on spoken QA, we applied the same data generation procedure to Spoken SQuAD paragraphs, generating 1K, 5K, and 10K question-answer pairs using Llama-3-8B, which achieved the best F1 performance. Interestingly, generating question-answer pairs from noisy ASR transcriptions is found to be competitive with using clean data, as no significant performance difference was observed. This may be due to the inherent noise introduced during synthetic data generation, even when using clean data.

4.2.2. Effect of Annotated Data Size on QA Performance

Previous experiments have shown that LLM-generated data effectively improve QA performance, especially when annotated data is scarce (e.g., 1K question-answer pairs), highlighting the importance of LLMs in low-resource scenarios. Here, we investigate whether this approach remains effective as the amount of annotated data increases. Figure 2 presents F1 scores for ALBERT models fine-tuned on 1K, 3K, and 5K annotated question-answer pairs, as well as 10K generated data for Spoken SQuAD. The 1K annotated data results were previously reported in Table 3. As shown in the figure, increasing the amount of annotated data reduces the gap between baseline models and those trained with additional generated data. This effect is especially pronounced for ALBERT-xxlarge, while for ALBERT-large, the improvements remain notable. These findings suggest that LLM-generated data plays a more critical role in low-resource scenarios, especially when using smaller models.

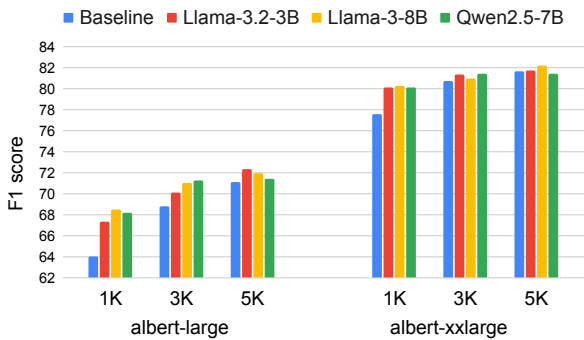


Figure 2: *F1 scores on Spoken SQuAD with 10K generated data for varying number of annotated QA pairs (1K, 3K and 5K).*

4.2.3. QA Performance on Noisy Spoken Data

Since the development set of Spoken SQuAD is available in three versions with varying levels of added noise (No Noise,

Table 4: *F1 results on noisy Spoken SQuAD development sets. WERs are also reported in parenthesis.*

	No Noise (22.7%)	Noise V1 (44.2%)	Noise V2 (54.8%)
Baseline (ALBERT-large)	64.0	45.4	33.4
+ 10K LLM data (Llama3-8B)	68.5	48.6	36.3
Baseline (ALBERT-xxlarge)	77.6	57.6	43.8
+ 10K LLM data (Llama3-8B)	80.3	60.7	47.2

Table 5: *F1 results on Turkish (text-based QA / spoken QA)*

	BERTurk	XLm-RoBERTa
Baseline	80.4 / 77.9	83.3 / 82.9
Llama-3.2-3B	81.9 / 78.2	86.6 / 83.2
Llama-3-8B	82.2 / 79.0	87.1 / 85.1
Qwen-2.5-7B	83.9 / 79.4	84.4 / 84.4

Noise V1, and Noise V2), we conducted experiments to evaluate performance on noisy spoken data. These experiments were specifically carried out for the 1K annotated subset and 10K generated data over SQuAD. The results are presented in Table 4. Despite a significant drop in F1 scores for noisy data with increasing WER, they still demonstrate the benefit of knowledge distillation from LLMs.

4.3. Turkish QA Results

For the baseline Turkish QA system, we fine-tuned the models described in Section 3.3 using the Turkish lecture QA dataset. The highest F1 score was achieved with the multilingual XLm-RoBERTa-large model (550M parameters). Additionally, we evaluated a smaller model, BERTurk (110M parameters). XLm-RoBERTa-large and BERTurk achieved F1 scores of 83.3 and 80.4 in text-based QA and 82.9 and 77.9 in spoken QA. Table 5 presents F1 results for the models fine-tuned with both LLM-generated and annotated data, with columns showing text-based and spoken QA results, respectively. The LLM-generated dataset for Turkish contains approximately 1.5K question-answer pairs. The results demonstrate that leveraging generated data is effective for both text-based and spoken QA in Turkish, even in limited-resource settings. Unlike English, Turkish is not officially supported by many LLMs. Despite the low answer relevancy scores reported in Table 1, achieving relative improvements of over 4% in text-based QA and over 2% in spoken QA is promising for low-resource languages and highlights the multilingual capabilities of off-the-shelf LLMs.

5. Conclusion

In this paper, we demonstrated that LLM-generated synthetic data significantly enhances the performance of QA models, particularly in low-resource settings. Our results confirm that even without fine-tuning, LLMs can effectively generate data to improve both text-based and spoken QA tasks, including for languages they do not officially support, highlighting their multilingual capabilities. Importantly, our approach leverages small, efficient models that remain viable alternatives in an era dominated by large-scale LLMs. Our findings reinforce the value of small models, as they offer computational efficiency while still benefiting from LLM-driven data generation. Future work can focus on further improving synthetic data quality and extending this approach to other low-resource languages beyond Turkish.

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